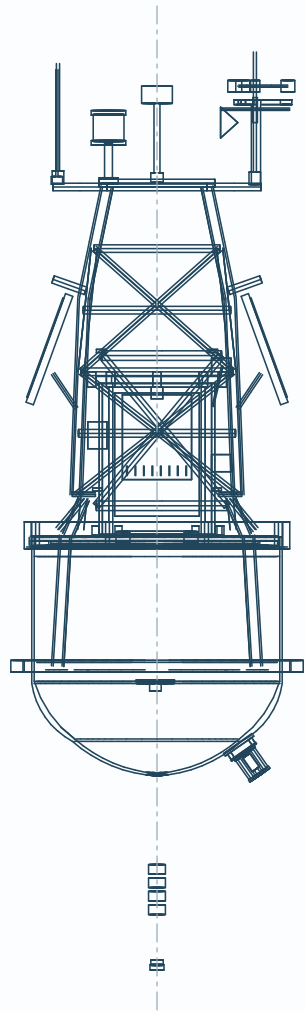


PROJECT FALCON-01 — COASTAL MONITORING BUOY

MASTER ASSEMBLY · SUBSYSTEMS · ELECTRONICS · POWER · MOORING · V2 CAD REFERENCE

REFERENCE / NOT FOR FABRICATION

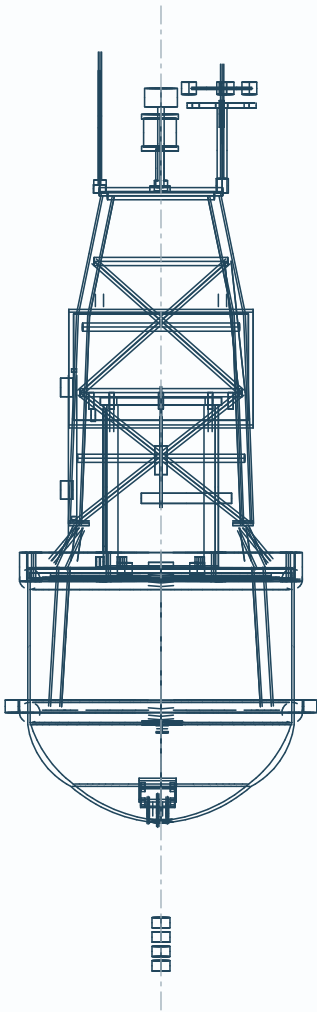
A — FRONT ELEVATION · CAD-PROJECTED



FLOAT Ø0.650 m · V2 CAD REF

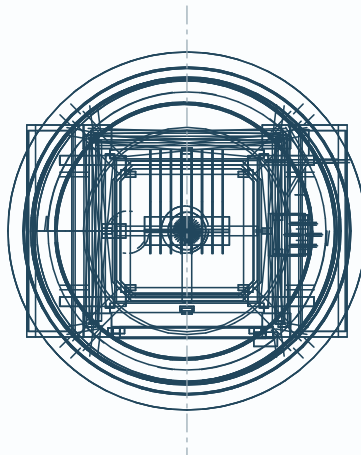
Overall installed height and loaded waterline: VERIFY IN NATIVE CAD / FLOTATION TEST

B — RIGHT ELEVATION

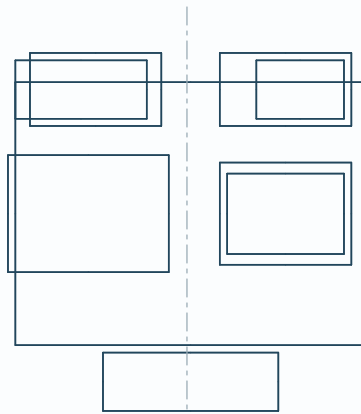


ORTHOGRAPHIC · DO NOT SCALE

C — PLAN VIEW

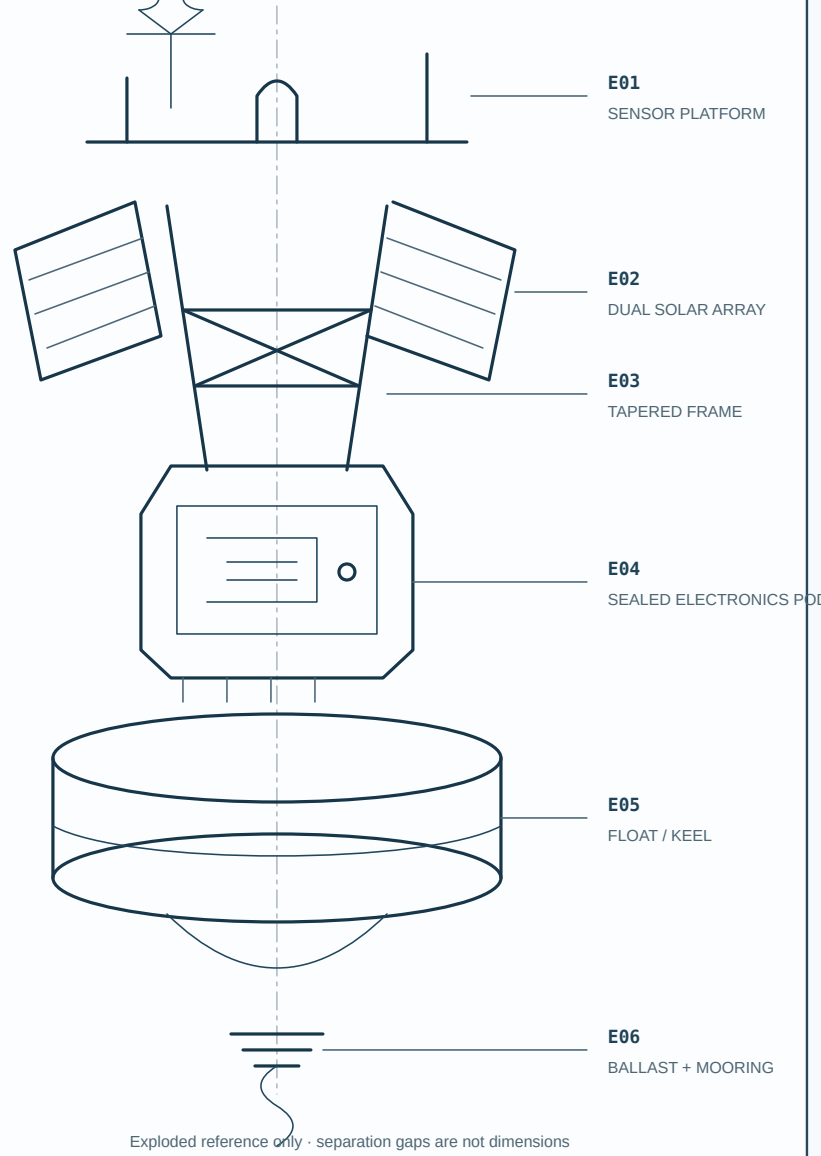


D — ELECTRONICS POD / EQUIPMENT PLAN



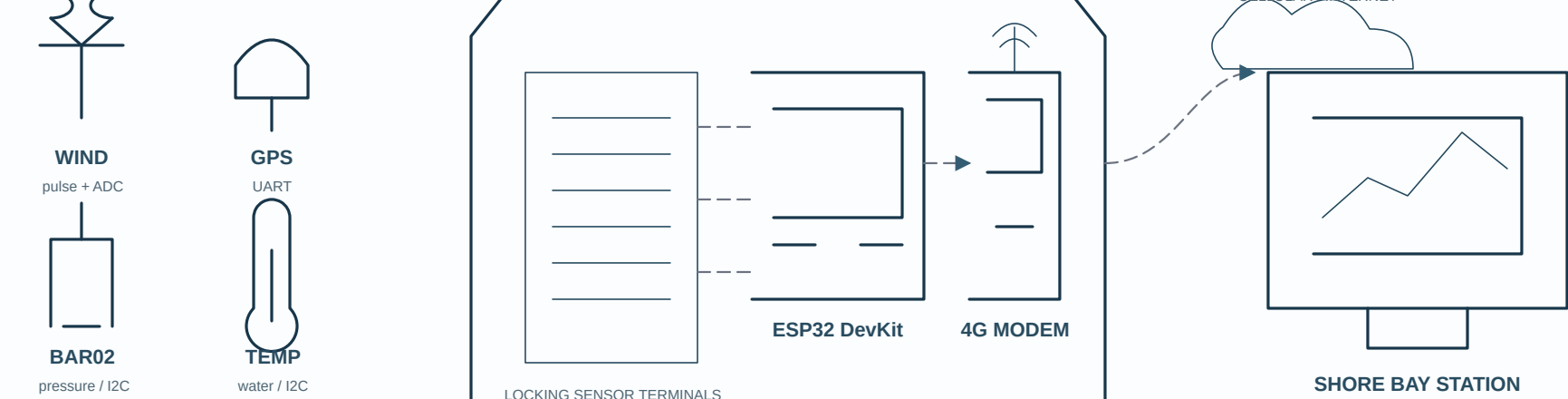
Orange Pi excluded · shore Bay Station only

E — EXPLODED ASSEMBLY VIEW · NTS



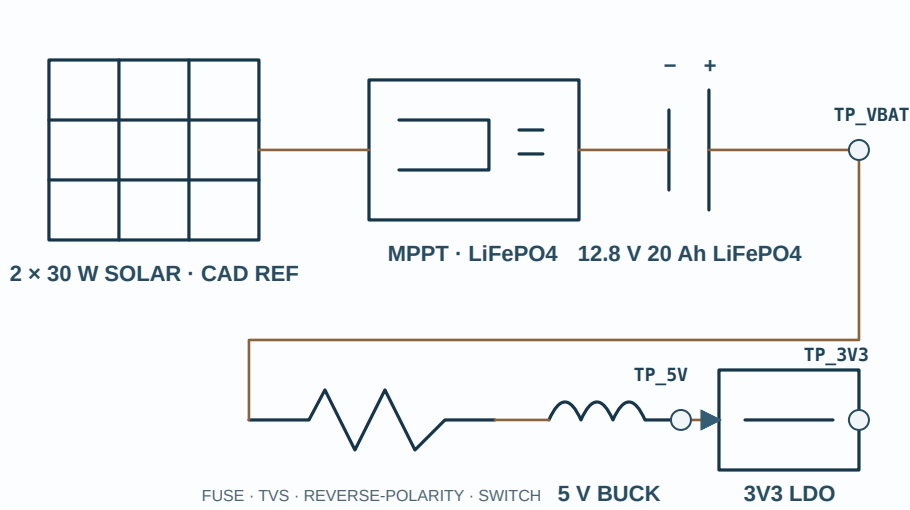
F — SUBSYSTEM INTERCONNECTION / DATA FLOW

SEALED ELECTRONICS POD



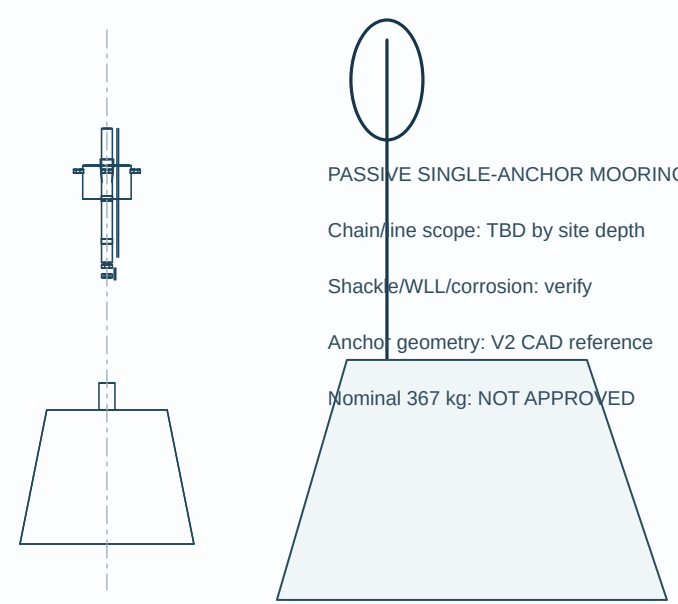
Sensor data → protected terminals → ESP32 validation/buffer → cellular telemetry → shore database, AI and dashboard

G — POWER MANAGEMENT



INA260 battery 0x40 · INA260 solar 0x41 · final protection ratings require measured load tests

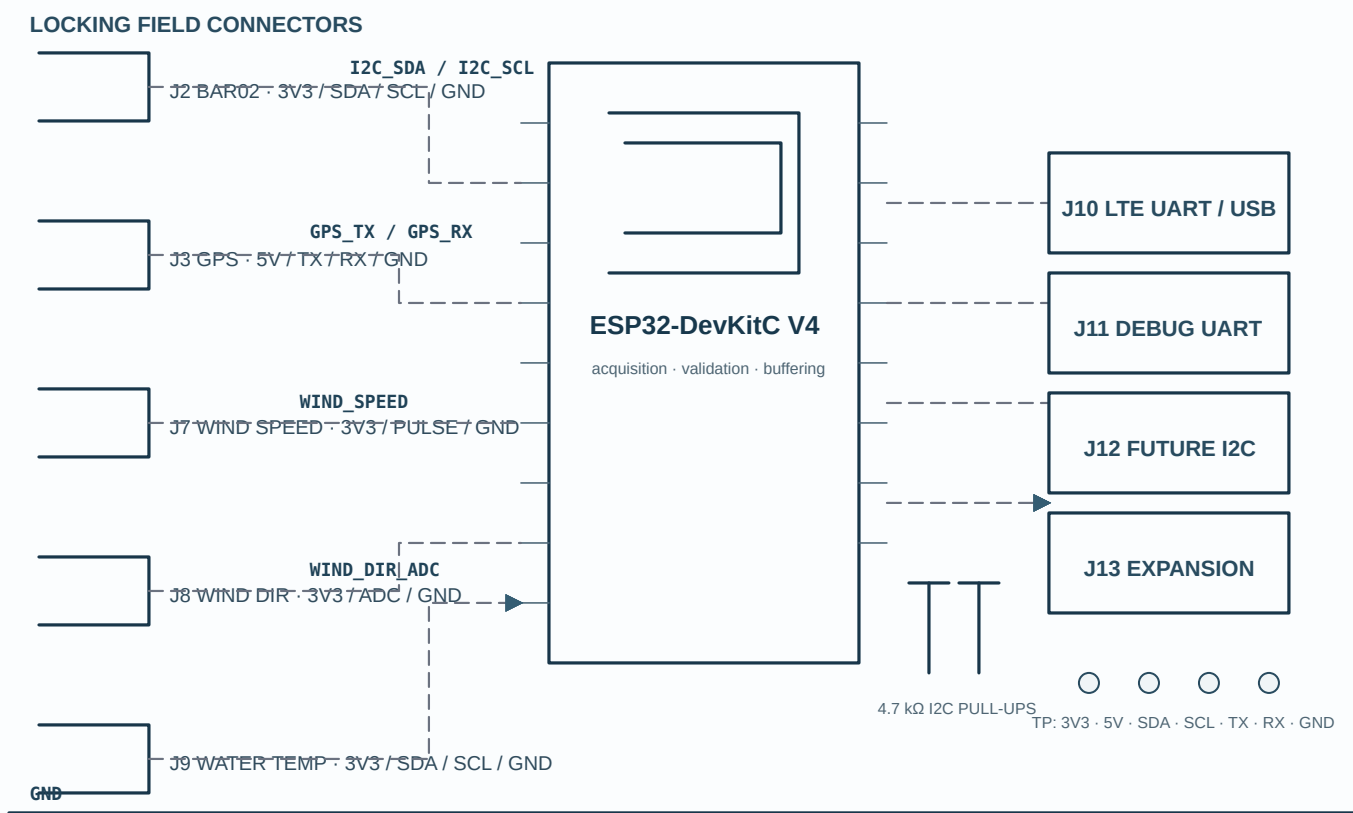
H — MOORING / ANCHORAGE



J — CONTROLLED COMPONENT REGISTER

ID	COMPONENT	STATUS	FUNCTION
S01	Blue Robotics Bar02 R2	PROTOTYPE	Pressure / wave input
S02	Adafruit Ultimate GPS PID 746	PROTOTYPE	Position / geofence
S03	SparkFun SEN-15901	PROTOTYPE	Wind speed / direction
S04	Blue Robotics Celsius R2	CANDIDATE	Water temperature
H01	Adafruit MCP9808 PID 1782	SELECTED	Enclosure temperature
H02	Adafruit INA260 PID 4226 x2	PROTOTYPE	Battery / solar health
C01	ESP32-DevKitC V4	SELECTED	Buoy controller
C02	SIM7600G-H 4G HAT	CANDIDATE	Cellular telemetry
P01	LiFePO4 12.8 V 20 Ah	RATING SET	Exact SKU / case TBD
SEC	Contact switch PID 375	PROTOTYPE	Enclosure-open state
A01	Buzzer + driver	TBD	Audible security alarm

K — SIMPLIFIED SENSOR / CONTROLLER WIRING



Functional wiring only—KiCad schematic controls pin numbers and manufacturing netlist · Orange Pi remains ashore

L — GENERAL NOTES

- Orthographic linework is projected from PROJECT-FALCON-V2.glb.
- V2 geometry remains a proposed replacement reference.
- Do not scale this drawing; verify native CAD and purchased parts.
- Dimensions, mass and placement marked TBD must not be invented.
- Complete stability, structure, ingress, thermal and mooring reviews.
- Orange Pi/Bay Station computer remains ashore.

PRELIMINARY ENGINEERING REFERENCE

M — DRAWING CONTROL

PROJECT	DRAWING TITLE
FALCON-01	MASTER ASSEMBLY & SYSTEMS
DRAWING NO.	FALCON-BP-000
SCALE	NTS
SOURCE	V2 GLB + DOCS
REVISION	P0
DATE	2026-09-01
STATUS	REFERENCE

RELEASE GATES / APPROVAL

- ☐ Native CAD dimension audit ☐ Purchased-part fit / connector clearance ☐ Loaded mass / displacement / waterline / freeboard
- ☐ CG / stability / righting test ☐ Structural load and fastener review ☐ IP / leak / thermal / corrosion validation
- ☐ Solar / battery measured load budget ☐ Pressure installation and wave-reference validation ☐ Mooring WLL / anchor design review

PREPARED BY / DATE

ENGINEERING REVIEW / DATE

THESIS ADVISER APPROVAL / DATE

NOT FOR FABRICATION